

Modelling with MATLAB Assignment 4

Contents

1	Question 1	1
1.1	Question 1a	1
1.2	Question 1b	1
1.3	Question 1c	2
2	Question 2	2
2.1	Question 2a	2
2.2	Question 2b	2
2.3	Question 2c	2
A	Figures	3

1 Question 1

Simple random walk in two dimensions.

1.1 Question 1a

I created the script file “q1script1.m” to simulate a simple random walk in two dimensions. First the file takes the number of steps required and a number of random walks. I chose to simulate 100 random walks. It then creates a matrix x to store the random walks in and another matrix dx that is randomly assigns directions. The directions are added to the x matrix. I used complex numbers to simulate walking in two-dimensional space with the real components being the x -coordinate and the imaginary components being the y -coordinates. From this I plotted the first two ‘people’s’ walks seen in 1.

1.2 Question 1b

The script file “q1script1.m” also works out the Euclidean distance of the walker, the mean of the Euclidean distance, the variance of the Euclidean distance and the probability that the person walks less than a distance of 10 from the origin.

To do this, first I worked out the Euclidean distance from the origin for each step of each persons walk

$$distance = \sqrt{x^2 + y^2}.$$

After that I found the mean of each of these distances and plotted them for each step taken. This can be seen in Figure 2. The graph shows that as the person takes more steps, they get further away from the origin.

I then found the variance of the Euclidean distance and plotted them for each step taken as seen in Figure 3. This graph shows that as the walkers take more steps, the variance between their distances from the origin increases.

Finally, the script “q1script1.m” approximates the probability that a walkers distance from the origin is less than 10 after 500 steps. I plotted the theoretical probability prediction of every step being less than 10 and the real data values of the steps being less than 10 on the same graph. The script also prints the values for the final point with the theoretical probability being 0.1816 and the real probability found to be 0.21.

1.3 Question 1c

I created a second script “q1script2.m” to work out the lowest number of steps to surpass a distance of 10 from the origin. I simulated 100 random walks. I made Figure 5 to plot the number of steps each individual needed to exceed a distance of 10 and also made a histogram to see the spread of the number of steps as seen in Figure 6. The script also prints the mean value which for my graphs was 107.23 steps. I added a line to Figure 5 at the mean distance.

2 Question 2

Stochastic logistic models.

2.1 Question 2a

Consider the stochastic logistic model:

$$dN = rN\left(1 - \frac{N}{K}\right)dt + \epsilon N dW, \quad N(0) = N_0, \quad t \in [0, T], \quad (1)$$

where N is the population density, r , K and ϵ are positive constants, W is a standard Brownian motion over $[0, T]$.

I created a script file “q2ascript.m” to simulate this equation using the Euler-Maruyama method. I used the conditions $r = 1, K = 1, T = 20, N_0 = 0.25$ and $\epsilon = 0.1$ and 0.3 . I plotted them on the same graph as seen in Figure 7. The population density rises quickly for both values of ϵ with $\epsilon = 0.1$ plateauing at around 1.1 and $\epsilon = 0.3$ plateauing at around 1.5.

2.2 Question 2b

Consider now the following modification of the above model:

$$dN = N\left(1 - \frac{N}{K}\right)(r dt + \epsilon dW), \quad N(0) = N_0, \quad t \in [0, T]. \quad (2)$$

I created a script file “q2bscript.m” to simulate this equation using the code for Question 2a as a base. The graphs created for Equation 2 rise quickly and then level off at population density 1.

2.3 Question 2c

I created a script file “q2cscript.m” to put graphs of Equation 1 and Equation 2 together to more easily analyse them. Clearly Equation 2 levels off to a smooth line whereas Equation 1 rises higher and does not become steady.

A Figures

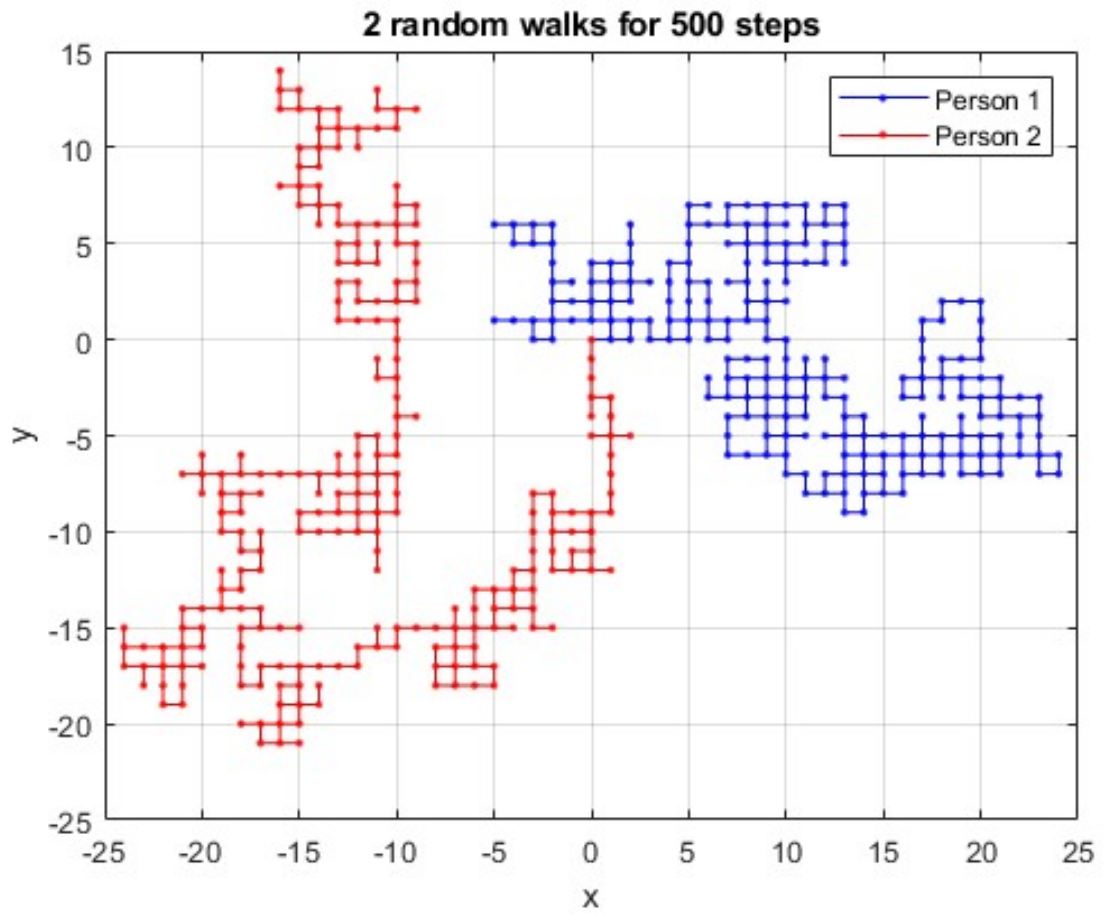


Figure 1: Graphical output of 2 random walks.



Figure 2: Graphical output of expected distance of random walks.

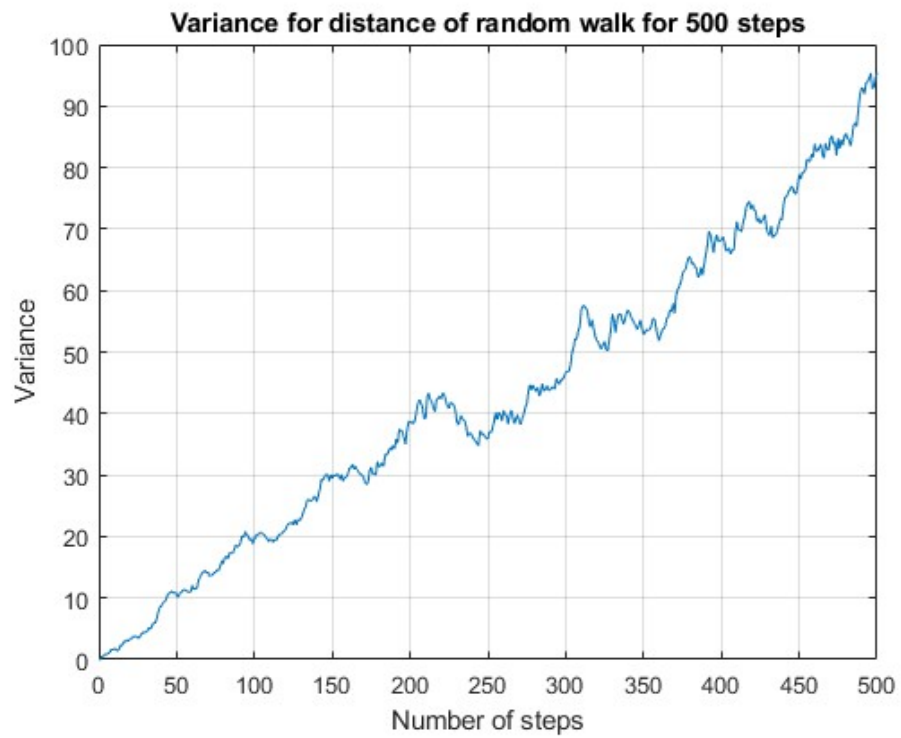


Figure 3: Graphical output of variance of random walks.

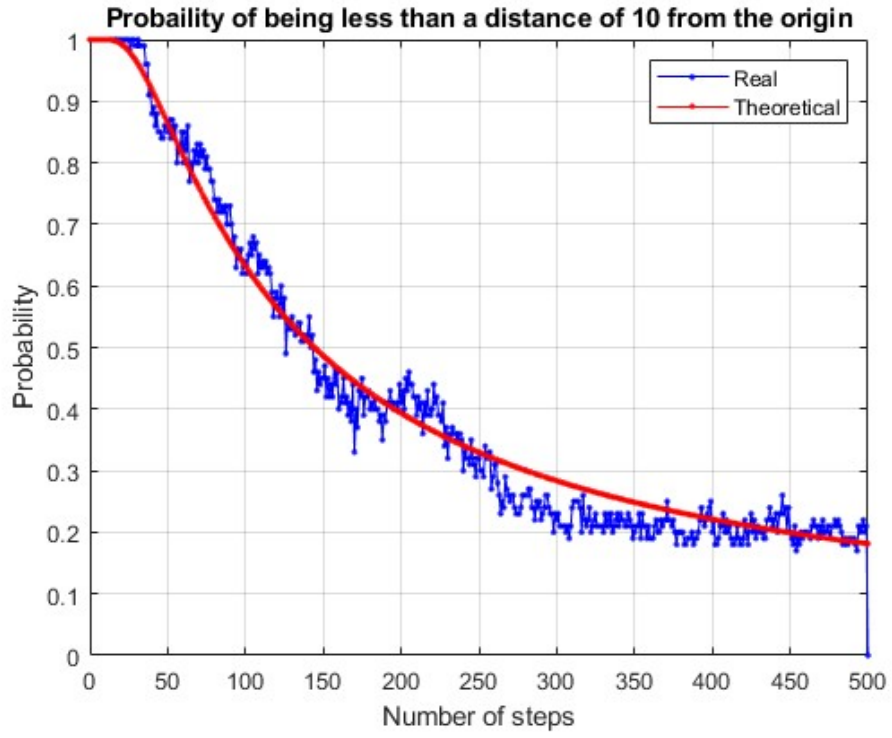


Figure 4: Graphical output of probability of being less than a distance of 10 from the origin.

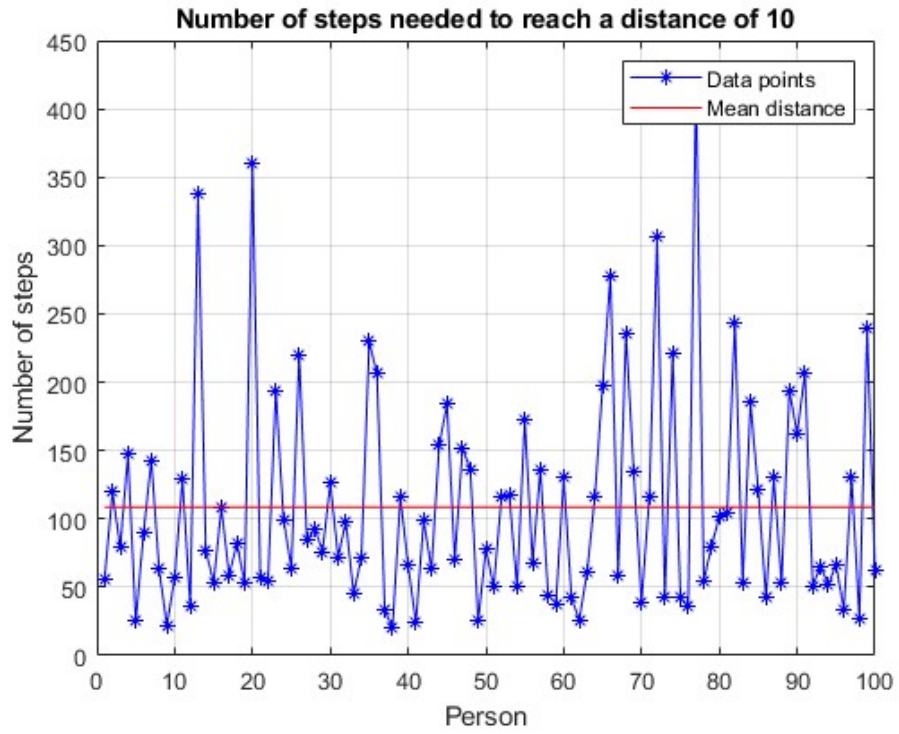


Figure 5: Graphical output of number of steps needed to reach a distance $L = 10$.

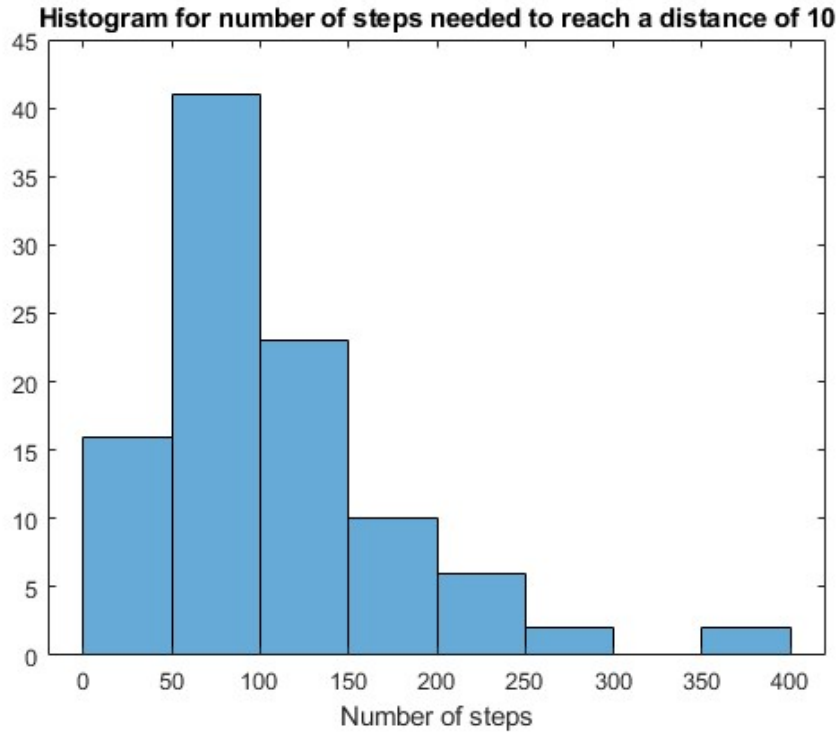


Figure 6: Histogram for the number of steps needed to reach a distance of $L = 10$.

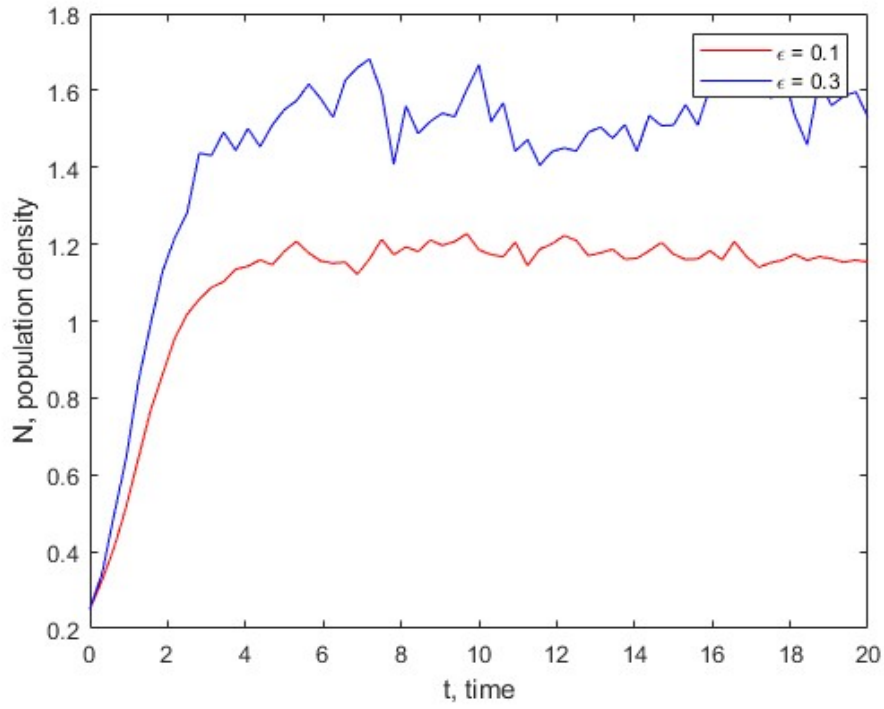


Figure 7: Simulations of Equation 1.

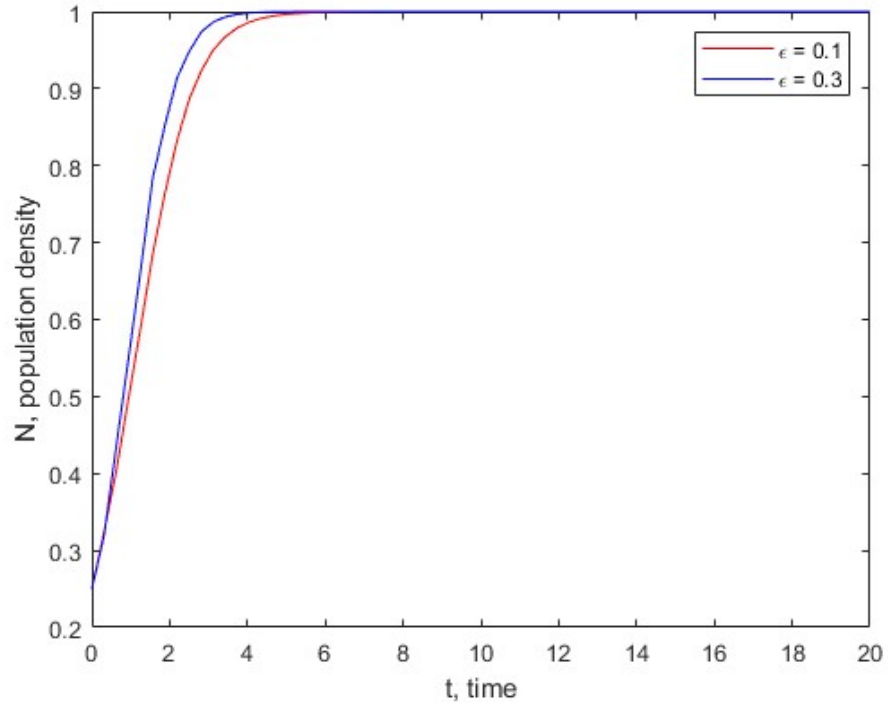


Figure 8: Simulations of Equation 2.

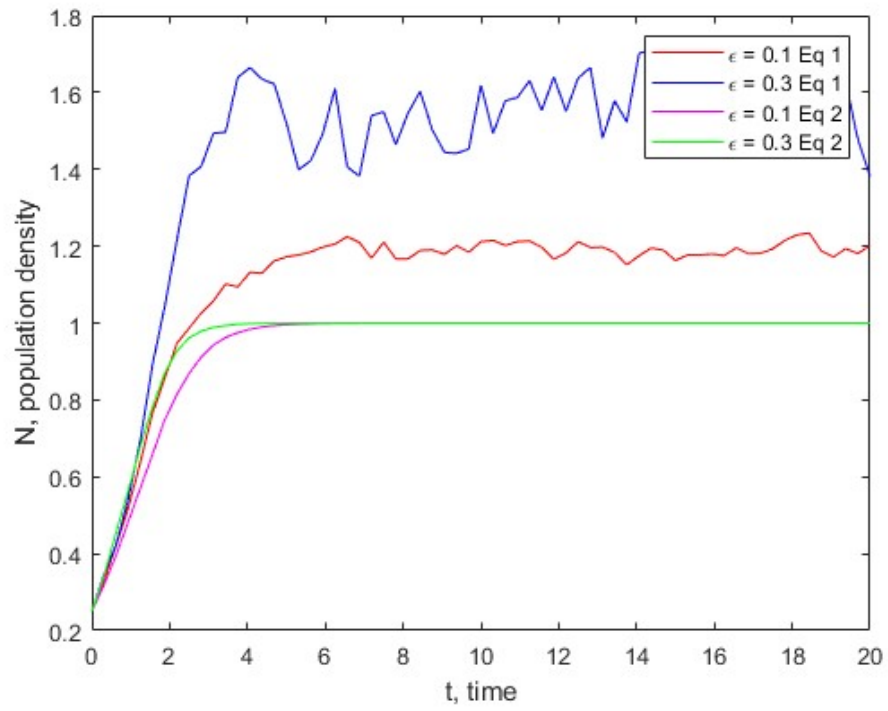


Figure 9: Simulations of Equation 1 and 2 together.